

12 **Ans : A**

To remain in contact at all times, acceleration of plate $\leq g$

$$\omega^2 x_0 \leq g \Rightarrow (2\pi f)^2 x \leq g$$

$$\text{maximum amplitude, } x_0 = 9.81 / (2\pi \times 20)^2 \Rightarrow x_0 = 6.2 \times 10^{-4} \text{ m}$$